

NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

PROBLEM SET 4

Output, the Interest Rate, and the Exchange Rate

1. Consider an open economy with flexible exchange rates. Suppose output is at the natural level, but there is a trade deficit. The goal of policy is to reduce the trade deficit and leave the level of output at its natural level. What is the appropriate fiscal and monetary policy mix?
2. A reduction in the interest rate in an economy operating under flexible exchange rates leads to an increase in output and a depreciation of domestic currency.
 - (a) How does the reduction in interest rates in an economy with flexible exchange rates affect consumption and investment?
 - (b) How does the reduction in interest rates in an economy with flexible exchange rates affect net exports?
3. Suppose there is an expansionary fiscal policy in the foreign country that increases Y^* and i^* at the same time.
 - (a) In an IS-LM-UIP diagram, show the effect of the increase in foreign output and the increase in the foreign interest rate on domestic output and the exchange rate, when the domestic central bank leaves the policy interest rate unchanged. Explain.
 - (b) In an IS-LM-UIP diagram, show the effect of the increase in foreign output and the increase in the foreign interest rate on domestic output and the exchange rate, when the domestic central bank matches the increase in the foreign interest rate with an equal increase in the domestic interest rate. Explain.
 - (c) In an IS-LM-UIP diagram, show the required domestic monetary policy following the increase in foreign output and the increase in the foreign interest rate, if the goal of the domestic monetary policy is to leave domestic output unchanged. Explain. When might such a policy be necessary?